

Programming Project / C++ – Bank Management System

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I. Task Description

Student 1 – responsible for:

- Entity, Client, Account, CurrentAccount, SavingsAccount modeling.
- writing objects to files
- cmdline parsing commands: Bank commands

Student 2 - responsible for:

- Bank, Transaction modeling.
- reading objects from files
- cmdline parsing commands: Client commands

II. Data structures

```
Bank{
    +vector<weak_ptr<Client>> clients
    +vector<unique_ptr<Account>> accounts
    +vector<unique_ptr<Transaction>> transactions
}
Entity{
    -id: int
}
Account{
    - id: int
    - owner: weak_ptr<Client>
    - iban: string
    - balance: double
    - currency: string
}
CurrentAccount: public Account{
    - managementFee
}
SavingsAccount: public Account{
    - interestRate
}
Client{
    -id: int
    -name: string
    +accounts: vector<weak_ptr<Accounts>>
    +bank: weak_ptr<Bank>
}
```

```
Transaction{
    - id: int
    - source: weak_ptr<Entity>
    - dest: weak_ptr<Entity>
    - amount: double
    - date: string/date
}
```

III. File Structure

accounts.txt [Draft]
id, ownerId, type, iban, balance, currency

clients.txt [Draft]
id, name

transactions.txt [Draft]
id, type, sourceId, destId, amount, date

IV. Interacting with executable

Application will offer the interaction between its components as follows:

`./bankapp.exe client -r -a -open <type> <currency>`

In order for a client to obtain an account of desired type and currency.

`./bankapp.exe client -r -a -close <iban>`

In order for a client to close a given account.

`./bankapp.exe client -d <iban> <amount>`

In order for a client to deposit an amount of currency in a given account.

`./bankapp.exe client -w <iban> <amount>`

In order for a client to withdraw an amount of currency from a given account.

`./bankapp.exe client -t <iban1> <iban2> <amount>`

In order for a client to transfer an amount of currency from account with **iban1** to account with **iban2**.

`./bankapp.exe client -q <iban1>`

In order for a client to query an account for its balance.

`./bankapp.exe bank -list -a [<clientId>]`

In order for bank to list all accounts of given client, or all bank's accounts if client parameter is missing.

```
./bankapp.exe bank -list -c
```

In order for bank to list all its clients.

```
./bankapp.exe bank -list -t <entityId>
```

In order for bank to list all its transactions.